$dN_p: T_pS \to T_pS$

自待 $dN_p(w_1), w_2$

$M \to S^2$

$T_pM \to T_{N(p)}S^2$ 线性映射

$\to dN_p: T_pS \to T_pS$

(w_1, w_2)

$\langle dN_p(w_1), w_2 \rangle = \langle w_1, dN_p(w_2) \rangle$

$X(u,v)$ $w_1 = X_u$ $w_2 = X_v$ $\alpha(t)$ $\alpha(0) = p$

$dN_p(\alpha'(0)) = \frac{d}{dt}(N(u(t), v(t))) = N_u \cdot u'(0) + N_v \cdot v'(0)$

$dN_p(X_u) = N_u$

$dN_p(X_v) = N_v$

$\langle N_u, X_v \rangle = \langle X_u, N_v \rangle$

$\langle N, X_u \rangle = 0 \Rightarrow \langle N, X_{uv} \rangle + \langle N_v, X_u \rangle = 0$

$\langle N, X_v \rangle = 0$

$Q(v) = \langle dN_p(v), v \rangle$

$\Rightarrow \mathrm{II}_p(v) = -Q(v)$

$\alpha(s)$ $\alpha(0) = p$. $N(s)$ $\langle N(s), \alpha'(s) \rangle = 0 \Rightarrow \langle N(s), \alpha'' \rangle = -\langle N', \alpha' \rangle$

$\alpha'(0) \to dN_p \to N'(0)$ 在0处 $\langle N(0), \alpha''(0) \rangle = -\langle dN_p(\alpha'), \alpha' \rangle$

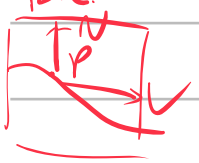
$|\alpha'| = k$.

(4) $\cos\theta = \langle n, N \rangle$

$k_n = k\cos\theta$

$\langle N(0), \alpha''(0) \rangle = \langle N, k_n \rangle = k\langle N, n \rangle = k\cos\theta = k_n$ → 法曲率

推论.



$z = y\varphi$ $(0,0,0)$ $dN_p = 0$ ← $k_n = 0$ ← $y(x) = x^4$ 在 $x=0$ $k=0$ $(x, x^4) \to (0, \varphi x^3)$

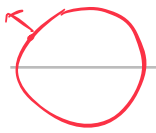
$\to (0, 12x^4)$

$= 0$



S^2 $I_p(v)$

$k = 1$.



↙ 法曲率 $|\kappa(-1)| = 1$ $I_p(v) = -1$

$dN_p \to T_pM$ e_1, e_2. $dN_p(e_1) = -k_1 e_1$

$dN_p(e_2) = -k_2 e_2$

$v = e_1\cos\theta + e_2\sin\theta$ $|v| = 1$

$k_n = -\langle dN_p(e_1\cos\theta + \dots), \dots \rangle$

$= k_1\cos^2\theta + k_2\sin^2\theta$

dN_p 行列式 $\det dN_p = K$

$\operatorname{trace} dN_p = -H$.

① 曲率线

脐点 $k_1 = k_2$

$X(u,v)$ $N=\frac{X_u \wedge X_v}{|X_u \wedge X_v|}$ $\alpha(t)$

$\alpha(t)$ 在 p 处 $\alpha' = X_u u' + X_v v'$

$dN_p(\alpha') = N_u u' + N_v v'$

$N_u = a_{11} X_u + a_{21} X_v$

$N_v = a_{12} X_u + a_{22} X_v$

$dN_p(\alpha') = (a_{11}u' + a_{12}v')X_u + (a_{21}u' + a_{22}v')X$

$dN_p\begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix}$

$e = \langle N, X_{uu} \rangle = -\langle N_u, X_u \rangle$

$f = \langle N, X_{uv} \rangle = _ _ .$

$g = \langle N, X_{vv} \rangle = _ _ _$

$-\begin{pmatrix} e & f \\ f & g \end{pmatrix} = A\begin{pmatrix} E & F \\ F & G \end{pmatrix}$. $A = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{pmatrix}$

$$A = -\begin{pmatrix} e & f \\ f & g \end{pmatrix}\begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

$a_{11} = \frac{fF - eG}{EG - F^2}$. $_ _$. $_ _ _$

$K = \det(a_{ij}) = \frac{eg - f^2}{EG - F^2}$

$-k_1, -k_2$ dN_p.

$dN + kI$ 不可逆. $k = k_1, k_2$

$\det\begin{pmatrix} a_{11}+k & a_{12} \\ a_{21} & a_{22}+k \end{pmatrix} = 0$ $k = k_1, k_2$.

$k^2 + k(a_{11} + a_{22}) + \det(a_{ij}) = 0$

$a_{11} + a_{22} = -(k_1 + k_2)$ $H = \frac{1}{2}(k_1 + k_2)$

$a_{11} = \frac{fF - eG}{EG - F^2}$ $a_{22} = _ _ .$ $H = \frac{1}{2}\frac{eG - 2fF + gE}{EG - F^2}$ ✓ $\Rightarrow k^2 - 2Hk + K^2 = 0$